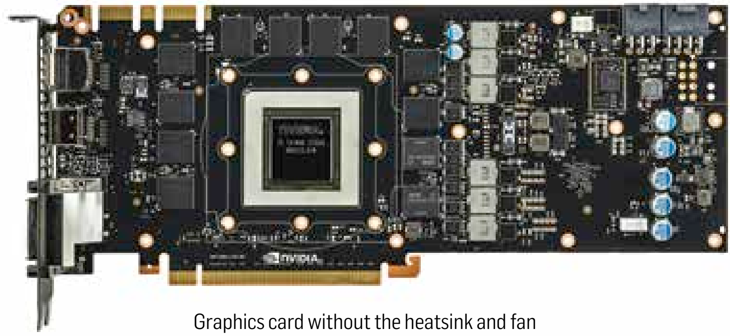
formats and bitmapping technologies to display pictures on the screen. But they are not well-equipped to deal with heavyweight 3D rendering. That's where a graphics card comes in.



Graphics card without the heatsink and fan

A discreet graphics card is a module that can process a lot of 3D information. Calculating vertices, their colour, tracing rays and their reflections, effects of shadows, dust and liquid in a scene are a few things that a graphics card can do really well. Along with this, calculation changes and stitching data together to form a complete scene is also done faster on a discreet graphics processing unit. The reason a graphics card works fast on these tasks can be attributed to three features of its architecture:

1. **Hardware for doing a task:** Many of the algorithms used for producing a 3D scene are not done by running mathematical instructions one after another on a general purpose processor (like your CPU). Instead, there are dedicated cores called streaming multiprocessors designed for such work. For example, Ray Tracing is typically done via hardware, than by software. You can read more about this on *https://en.wikipedia.org/wiki/Ray-tracing_hardware* (if you want to get into details, study the references at the bottom of the article). A graphics card packs many such dedicated streaming multiprocessors to enable faster graphics data processing.
2. **Parallelism by using very high number of processors:** The CPU is a general purpose processor that can run billions of instructions per second. A graphics processor on the other hard is dumb. It cannot do scheduling, memory management, intelligent interrupt handling etc. However, they have a large number of registers available and the number of processors is high and they work at a very fast rate. For example, the Nvidia GTX 1080 Ti has 3500+ processors, each of which can run at 1.5